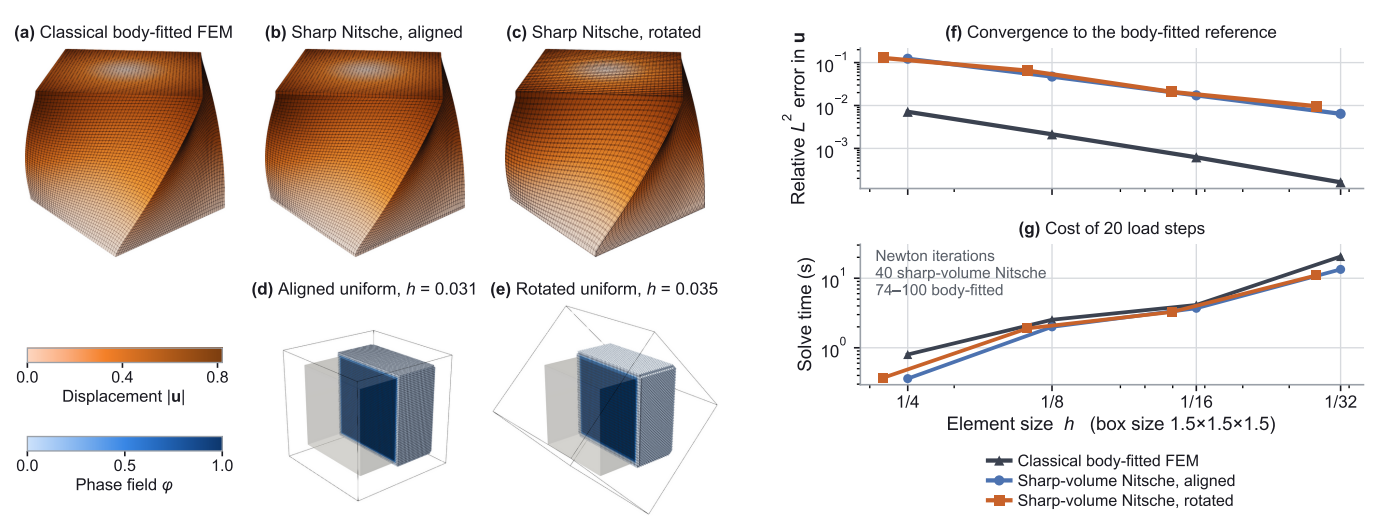

(a) Classical body-fitted FEM
(b) Sharp Nitsche, aligned
(c) Sharp Nitsche, rotated
(d) Aligned uniform, h = 0.031
(e) Rotated uniform, h = 0.035
0.0
0.4
0.8
Displacement |u|
0.0
0.5
1.0
Phase field φ
(f) Convergence to the body-fitted reference
Relative L² error in u
10⁻¹
10⁻²
10⁻³
(g) Cost of 20 load steps
Newton iterations
40 sharp-volume Nitsche
74–100 body-fitted
Solve time (s)
10¹
10⁰
1/4
1/8
1/16
1/32
Element size h (box size 1.5×1.5×1.5)
Classical body-fitted FEM
Sharp-volume Nitsche, aligned
Sharp-volume Nitsche, rotated